



WACHEMO UNIVERSITY
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERING

**PROJECT TITLE: WACHEMO UNIVERSITY
TRANSPORTATION SYSTEM**

GROUP 9

NAME

ID

1. ETSUBDINK GETACHEW	WCU173021
2. BETELIHEM ZEKEIOS	WCU174940
3. BIRUKTAWIT AREDA	WCU172814
4. NATINAEL TEKILE	WCU17D0815

PROGRESS SUBMISSION DATE: 27/03/2018 E.C

Acknowledgement

First and foremost, we thanks Almighty God for granting us the strength, knowledge, and perseverance to complete this WACHEMO UNIVERSITY TRANSPORTATION SYSTEM project successfully. We also extend our sincere gratitude to our instructor.

Contents

Chapter One: Introduction.....

1.1. Background of the Organization.....	
1.2. Statement of the Problem.....	
1.3. Objectives of the project.....	
1.3.1. General objective.....	
1.3.2. Specific objective(s).....	
1.4. Methodologies	
1.4.1. Requirement Elicitation Methodology	
1.4.2. Requirement analysis and Modeling	
1.4.3. System Implementation Methods	
1.5. Scope and Limitation of Project.....	
1.5.1. Scope.....	
1.5.2. Limitation.....	
1.6. Significance and Beneficiariesof the project.....	
1.7. Feasibility analysis.....	
1.7.1. Operational/Organizational Feasibility.....	
1.7.2. Technical Feasibility.....	
1.7.3. Economic Feasibility.....	
1.7.4. Schedule Feasibility	
1.7.5. Legal Feasibility.....	
1.8. Risk Assessment	

1.9. Development Tools.....	
1.10. Work Breakdown	
1.10.1. Project plan Activities (Schedule)	
1.10.2. Project Organization	
1.10.3. Team Organization	
1.11. Budget plan.....	

Chapter Two: Requirement Analysis and Specifications.....

2.1. Description of the Current System.....	
2.2. Players/Actors in the Existing System.....	
2.3. Use case Diagram for Existing System	
2.4. Forms and Documents used in the Existing System.....	
2.5. Business Rule of Existing System.....	
2.6. Proposed System	
2.6.1. Overview	
2.6.2. Business Rule of the New System	
2.6.3. Functional Requirements	
2.6.4. Non-Functional Requirements	
2.6.5. Actor and Use case Identification	
2.6.6. System Model	
2.6.6.1. Use case Model	
2.6.6.1.1. Use Case Description	
2.6.6.2. Sequence Diagram	
2.6.6.3. Activity Diagram	
2.6.6.4. State chart Diagram	

2.6.6.5.	Class Diagram	
2.6.6.6.	User Interface Prototyping	
Chapter Three: Database Design.....		
3.1.	Conceptual Database Design of the New System.....	
3.1.1.	Entities identification and Description.....	
3.1.2.	Attributes identification and description.....	
3.1.3.	Relationship identification and description.....	
3.1.4.	ER Diagrams.....	
3.2.	Logical Database Design of the New System.....	
3.2.1.	ER to Table Mapping.....	
3.2.2.	Validate Model using Normalization	
3.2.2.1.	First Normal Form (1NF).....	
3.2.2.2.	Second Normal Form (2NF).....	
3.2.2.3.	Third Normal form (3NF)	
3.2.2.4.	Other NF.....	
3.2.3.	Relational Schema with Referential Integrity after normalization.....	
3.3.	Physical Database Design of the New System.....	
3.3.1.	Physical Design Strategy.....	
3.3.2.	Hardware Implementation.....	
Chapter Four: System Design.....		
4.1.	Introduction.....	
4.2.	Purpose of the System.....	
4.3.	Design Goals.....	
4.4.	Current Software Architecture.....	
4.5.	Proposed Software Architecture.....	
4.5.1.	Subsystem Decomposition.....	
4.5.2.	Component Diagram.....	
4.5.3.	Deployment Diagram.....	
4.5.4.	Database Diagram.....	

4.5.5. Persistent Data Management.....	
4.5.6. Access Control and Security.....	
4.5.7. Global Software Control.....	
4.5.8. Boundary Conditions	
chapter Five: Implementation and Testing.....	
5.1. Test Procedure	
5.2. User Manual Preparation	
.....	
5.3. Training and Installation.....	
5.4. Startup Strategy	
Chapter Six: Conclusion and Recommendation.....	
6.1. Conclusion.....	
6.2. Recommendation.....	
6.3. Reference.....	
6.4. Appendix	

CHAPTER ONE

INTRODUCTION

1.1 Background of the Organization

Wachemo University is a public higher education institution located in Hossana, Ethiopia, serving thousands of students, faculty, and staff across two campuses. Efficient transportation is essential to ensure students and staff can move between classrooms(hospital), offices, and other university facilities. Currently, transportation operations rely on manual scheduling and communication, which causes delays, confusion, and inefficient use of vehicles. A centralized transportation management system is needed to streamline vehicle scheduling, tracking, and usage reporting, improving the overall campus experience.

1.2 Statement of the Problem

The absence of a structured transportation system at Wachemo University

leads to several issues:

- 1.Students and staff face delays due to inefficient scheduling of vehicles.
- 2.Tracking vehicle usage and maintenance is difficult and prone to errors.
- 3.Communication between transportation staff and users is inconsistent.
- 4.Manual management increases workload for the transportation system.

1.3 Objectives of the Project

1.3.1 General Objective

To develop a centralized and efficient transportation management system for Wachemo University.

1.3.2 Specific Objectives

- 1.To enable students and staff to request and book transportation online.
- 2.To assign vehicles and drivers efficiently based on demand and availability.
- 3.To track vehicle maintenance, fuel consumption, and usage history.
- 4.To generate reports for university management to optimize transport operations.

1.4 Methodologies

1.4.1 Requirement Elicitation Methodology

- 1.Conduct interviews with transportation staff and university administrators.
- 2.Distribute questionnaires to students and faculty to understand their transportation needs.
- 3.Observe current transportation operations to identify gaps.

1.4.2 Requirement Analysis and Modeling

Develop use case diagrams to represent the system's functions and actors. Create activity diagrams to visualize the flow of transportation processes. Identify functional and non-functional requirements based on gathered data.

1.4.3 System Implementation Methods

Develop a web-based application for accessibility on multiple devices.
Use a relational database (e.g. SQL) to store vehicle, driver, and user data.
Implement reporting and notification features to support decision-making and user communication.

1.5 Scope and Limitation of the Project

1.5.1 Scope

Centralized management of all university-owned vehicles.
Online booking and scheduling system for students and staff.
Tracking of vehicle assignment, usage, and maintenance.
Generation of reports for transportation management.

1.5.2 Limitation

System only covers university-owned vehicles.
Real-time GPS tracking may be limited by available hardware.
Internet access is required for remote use of the system.

1.6 Significance and Beneficiaries

Students and staff: Can easily book and track transportation services.
Transportation management: Can monitor vehicle usage and maintain records efficiently.
University administration: Gains insight into transportation operations, enabling informed decision-making.

1.7 feasibility analysis

1.7.1 Operational Feasibility

The system aligns with the university's transportation operations and user needs.

1.7.2 Technical Feasibility

The university has adequate IT infrastructure and skilled personnel to support a web-based system.

1.7.3 Economic Feasibility

The benefits of improved efficiency and reduced manual workload outweigh the system development costs.

1.7.4 Schedule Feasibility

The project can be completed within the semester, following a structured plan.

1.7.5 Legal Feasibility

The system will comply with university policies, data privacy, and software licensing rules.

1.8 Risk Assessment

- 1) Hardware or server failure could disrupt system operation.
- 2) Users may resist adopting a new digital system.
- 3) Data loss due to improper backups or technical errors.
- 4) Limited budget could affect development scope.

1.9 Development Tools

Programming Language: computer programming(C++)

Database : SQL server

IDE: Visual Studio, enterprise architecture

Other Tools: Diagram tools

1.10 Work Breakdown

1.10.1 Project Plan Activities

Activities	Time
Requirement gathering	1 weeks
Requirement analysis and modeling	2 weeks
System design	
Data base design	
Implementation and testing	
Deployment and user training	

1.10.2 Project Organization

- * Project Manager: Oversees progress
- * Developers: Implement system
- * Database Designer: Designs and manages database
- * Tester: Ensures system works correctly
- * Documentation Specialist: Prepares manuals and reports

1.10.3 Team Organization

Group members	Task
Biruktawit Arede	System designer
Betelihem Zekios	Requirement gathering and documentation
Natinael Tekile	Implementation and coding
Etsubdink Getachew	Project manager and testing and

	organizing
--	------------

1.11 Budget Plan

Hard Ware material	Quantity	Estimated cost
Laptop	1	35,000
USB cable		
Pen	3	100
Flash disk	–	
For print	–	
Total		

CHAPTER TWO

Requirement Analysis and Specifications

2.1 Description of the Current System

Right now, the transportation service at Wachemo University works manually. Students and staff usually request vehicles by:

- 1) Going physically to the transportation office

- 2) Calling the office or drivers, or
- 3) Sending informal messages.

Vehicle assignments are written on paper or in simple notebooks. There is no centralized platform that records bookings, vehicle availability, driver schedules, fuel usage, or maintenance history. Because of this, the transportation office struggles to keep track of daily requests, and users cannot see the status of their booking.

Currently, there are about 45 cars at Wachamo University, and each one is controlled and managed using a paper-based system. And this has different challenges and problems.

2.2 Players/Actors in the Existing System

These are the people involved in the current manual process:

1. Transportation Manager – Assigns vehicles and drivers manually.
2. Drivers – Take instructions directly from the manager.
3. Students/Staff (Users) – Request transportation physically or by phone.
4. Mechanic/Maintenance Staff – Informed manually about vehicle issues.

2.3 Use Case Diagram for Existing System (Manual System)

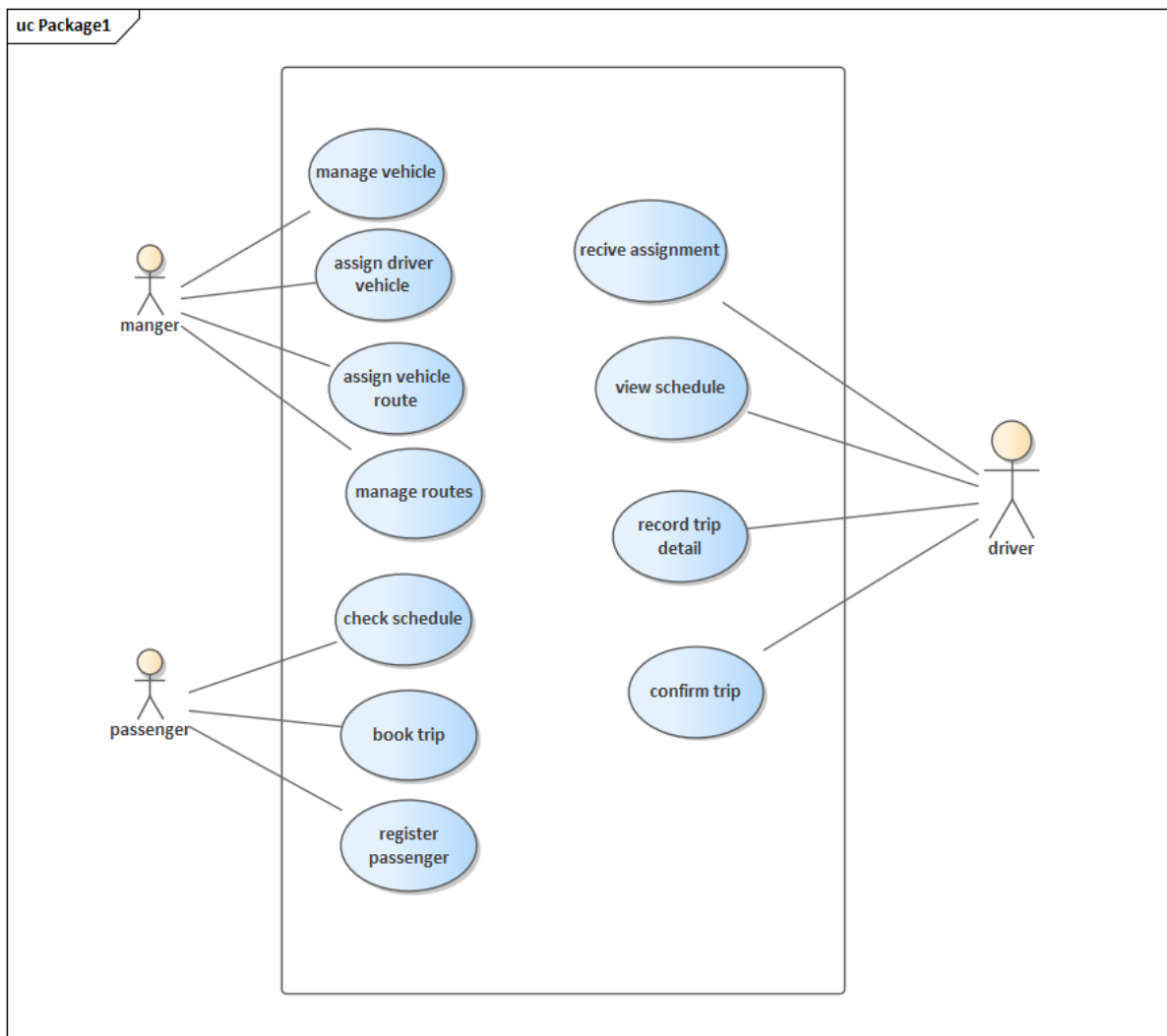


Figure 1.1 Use Case Diagram

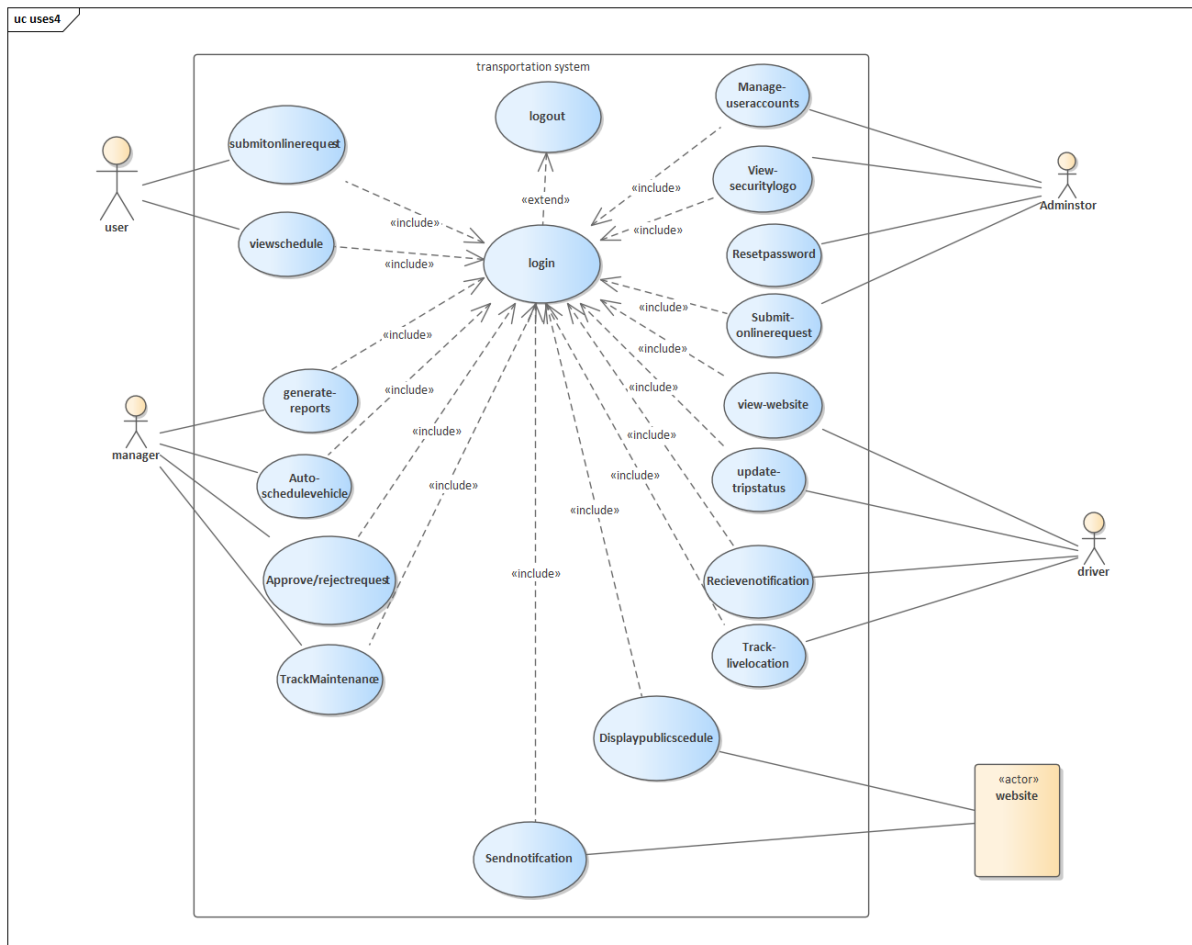


Figure 2.2 Use Case Diagram For New System

2.4 Forms and Documents Used in the Existing System

Currently, the transportation department uses:

Vehicle request forms (paper-based)

Driver assignment sheets

Vehicle logbooks for tracking trips

Maintenance request form

fuel consumption

Replacing car wheels (paper based)

All of these are stored physically and can easily get lost or damaged.

2.5 Business Rule of Existing System

Business rules are decisions or policies that guide how the current system works.

Examples:

Only the transportation manager can approve vehicle requests.

Drivers cannot take a trip without assignment from the manager.
Ministry vehicles must be recorded in the logbook after each trip.
Requests are first-come, first-served unless urgent.

2.6 Proposed System

2.6.1 Overview

The proposed system is a web-based platform that allows students and staff to request transportation online. The transportation manager can assign vehicles and drivers through the system, monitor availability, and track maintenance. Drivers can view their assignments digitally. Reports will be generated automatically.

2.6.2 Business Rules of the New System

These rules guide the new system:

1. Only registered users can request transportation online.
2. The system automatically checks vehicle availability before approval.
3. Transport manager must approve or reject each request.
4. Drivers receive assignments through the system.
5. Vehicle records (trips, fuel use, maintenance) must be updated after every trip.
6. Only admins can manage vehicles, drivers, and system settings.

2.6.3 Functional Requirements

These are the system features — what the system must do.

1. User Registration & Login

Allow users to create accounts and sign in.

2. Request Transportation

Users can request a vehicle by selecting date, time, and destination.

3. Approve or Reject Requests

Admin or transport manager reviews requests.

4. Vehicle & Driver Management

Add, update, or remove vehicles and drivers.

5. Assign Driver & Vehicle

Manager assigns the driver and vehicle to the request.

6. Notification System

Notify users when their request is approved.

7. Trip Record Management

Drivers record trip details after completion.

8. Maintenance Management

Track repairs, service dates, and fuel records.

9. Generate Reports

Monthly and weekly reports for management.

2.6.4 Non-Functional Requirements

It describes how system quantity should behave .

1. Usability – Easy to use for students and staff.
2. Performance – System should handle multiple requests efficiently.
3. Security – Password protection, role-based access control.
4. Reliability – System should operate without frequent failures.
5. Scalability – Able to handle more users in the future.
6. Availability – System should be accessible as long as internet is available.

2.6.5 Actor and Use Case Identification

Actors in the new system:

User (Student or Staff)

Transport Manager

Driver

Administrator

Main Use Cases:

Register/Login

Request vehicle

Approve request

Assign driver & vehicle

Update vehicle info

Manage drivers

View assignments

Record trip details

Generate reports

2.6.6 System Models

2.6.6.1 Use Case Model

A diagram showing actors and their interactions with the system.

2.6.6.1.1 Use Case Descriptions

For example:

Use Case: Request Transportation

Actor: Student/Staff

Pre-condition: User must be logged in

Main Flow:

1. User selects request form
2. Fills in trip details
3. System saves request

Post Condition: Request sent to manager for approval

2.6.6.2 Sequence Diagram

Shows step-by-step interaction between user and system.

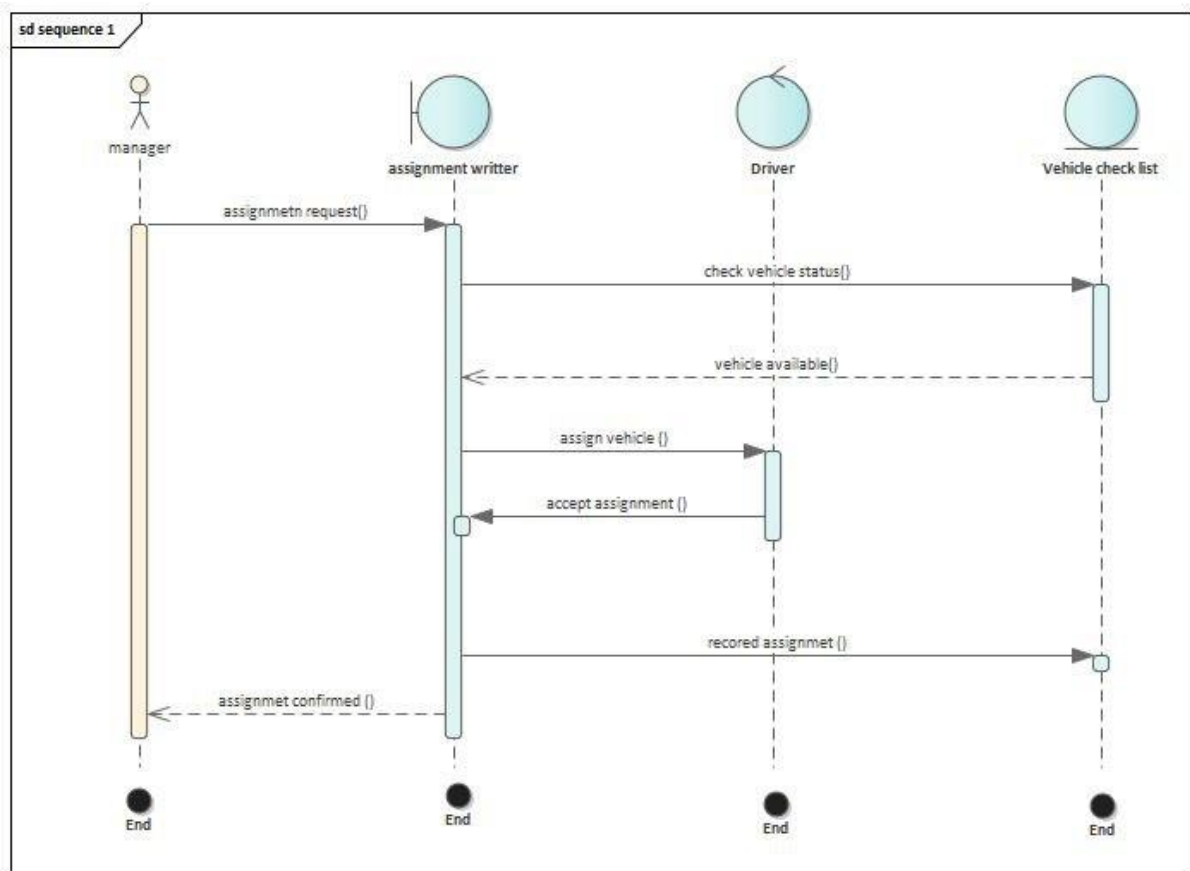


Figure 2.3 sequence diagram(assign driver to vehicle)

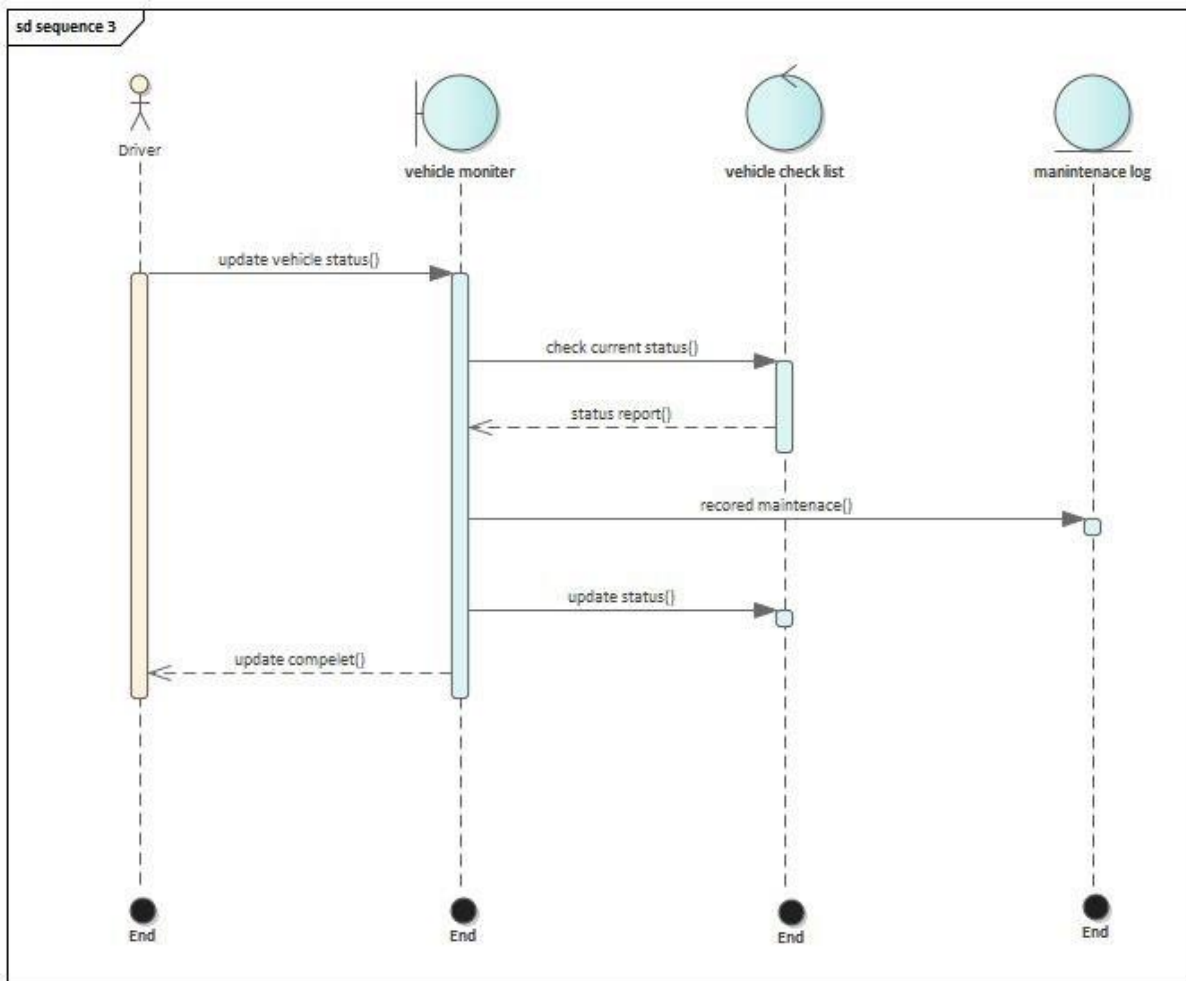


Figure 2.4 Sequence Diagram(manage vehicle)

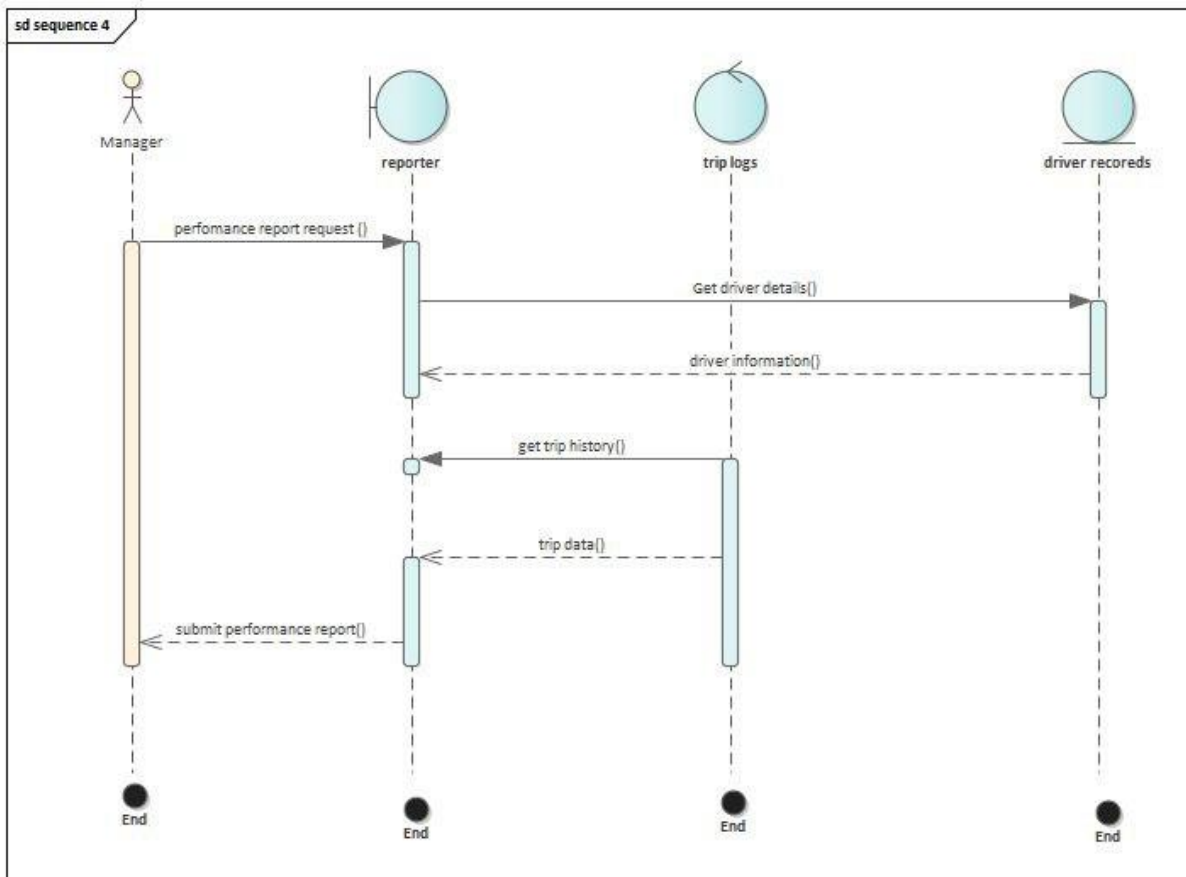


Figure 2.5 Sequence Diagram(monitored driving performance)

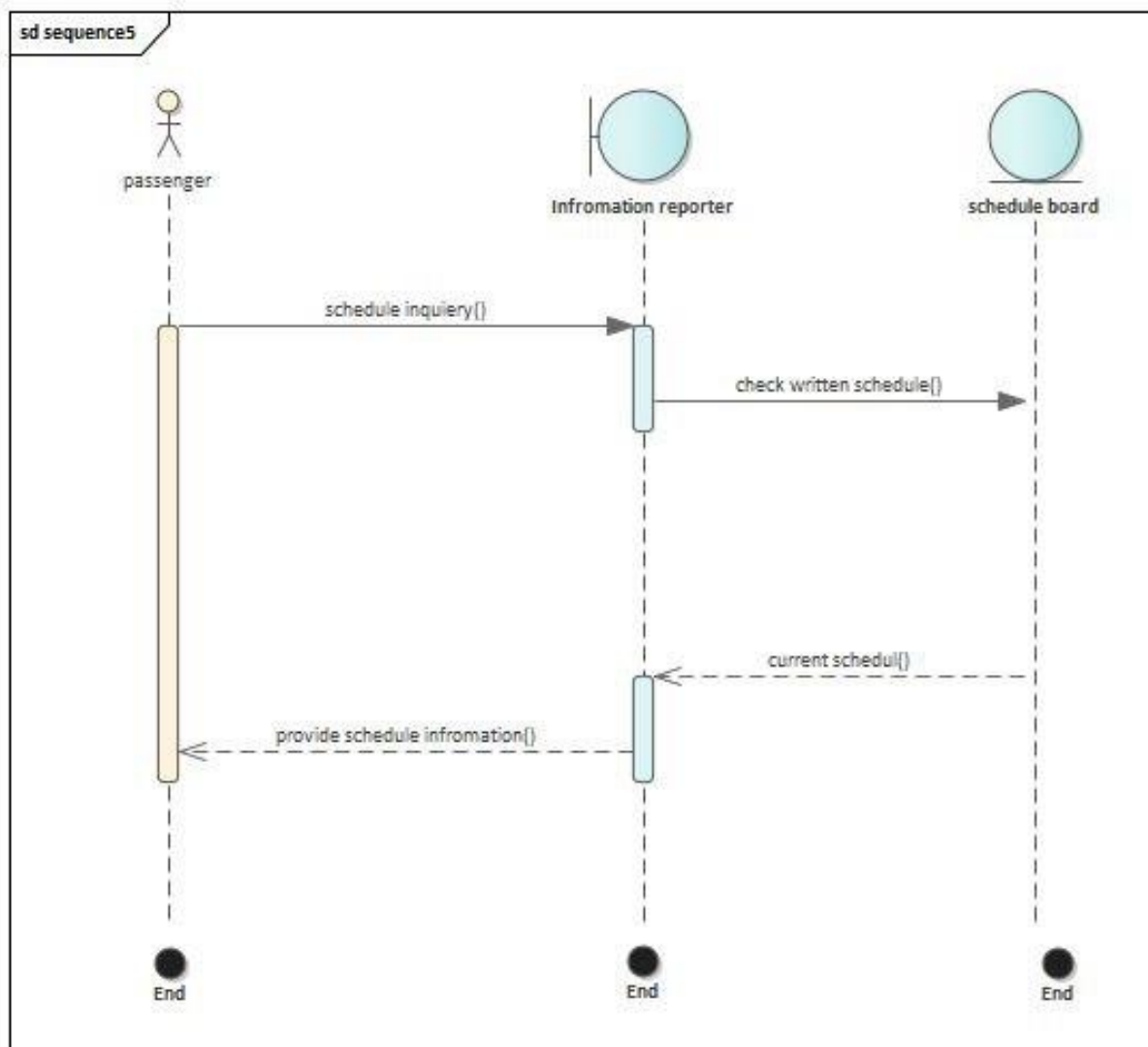


Figure 2.6 Sequence Diagram(view schedule)

2.6.6.3 Activity Diagram

Shows workflow for requesting and approving transport.

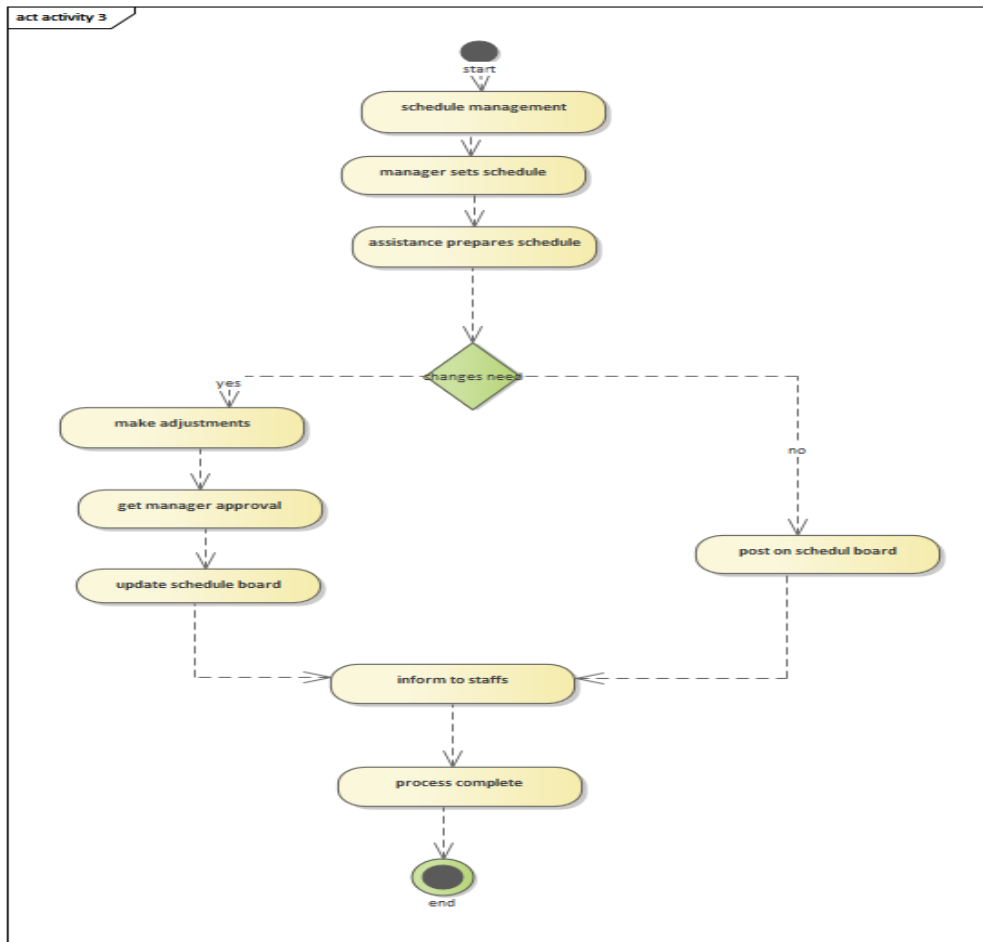


Figure 2.7 Activity Diagram(schedule management flow)

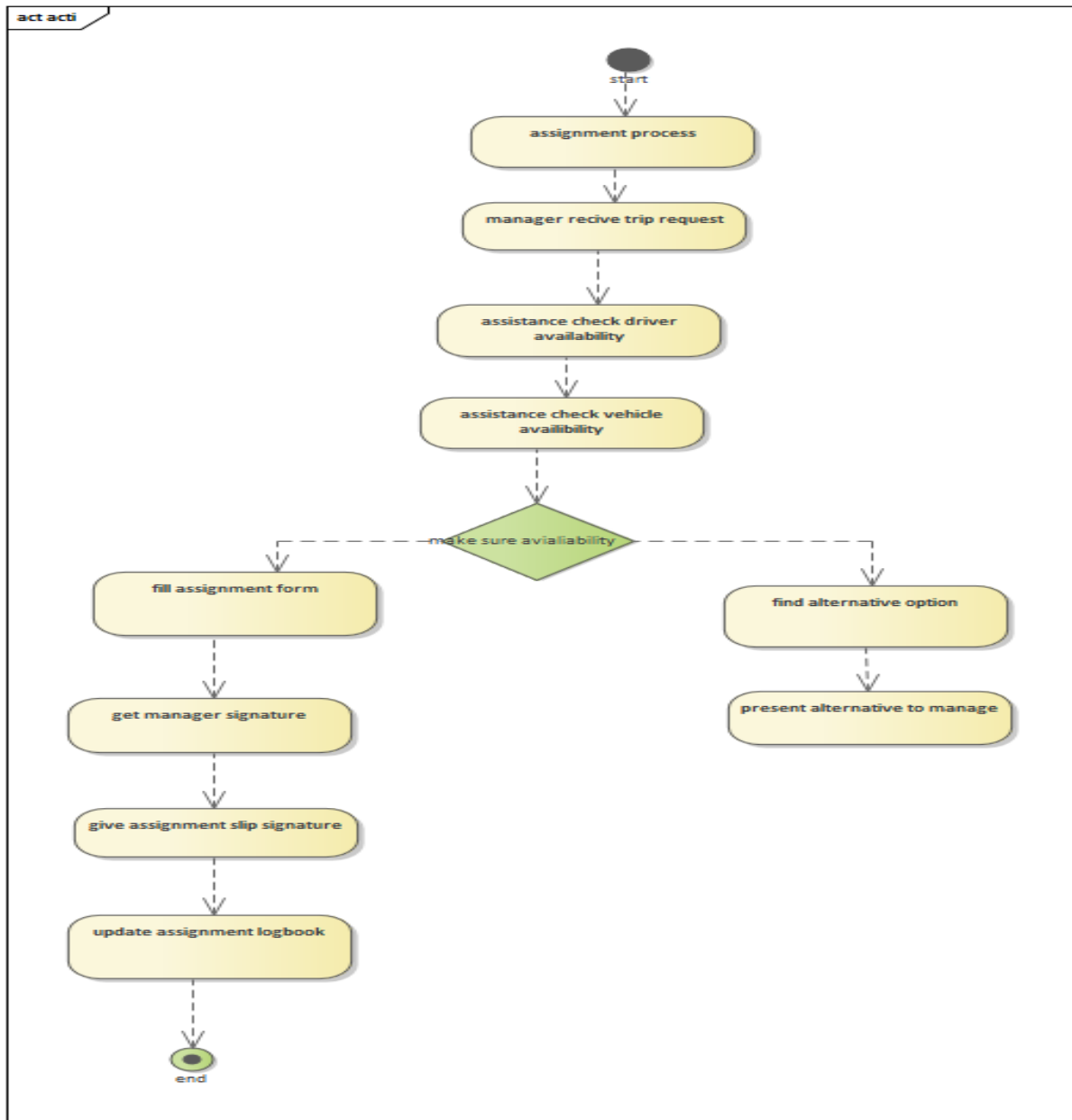


Figure 2.8 Activity Diagram(driver assignment flow)

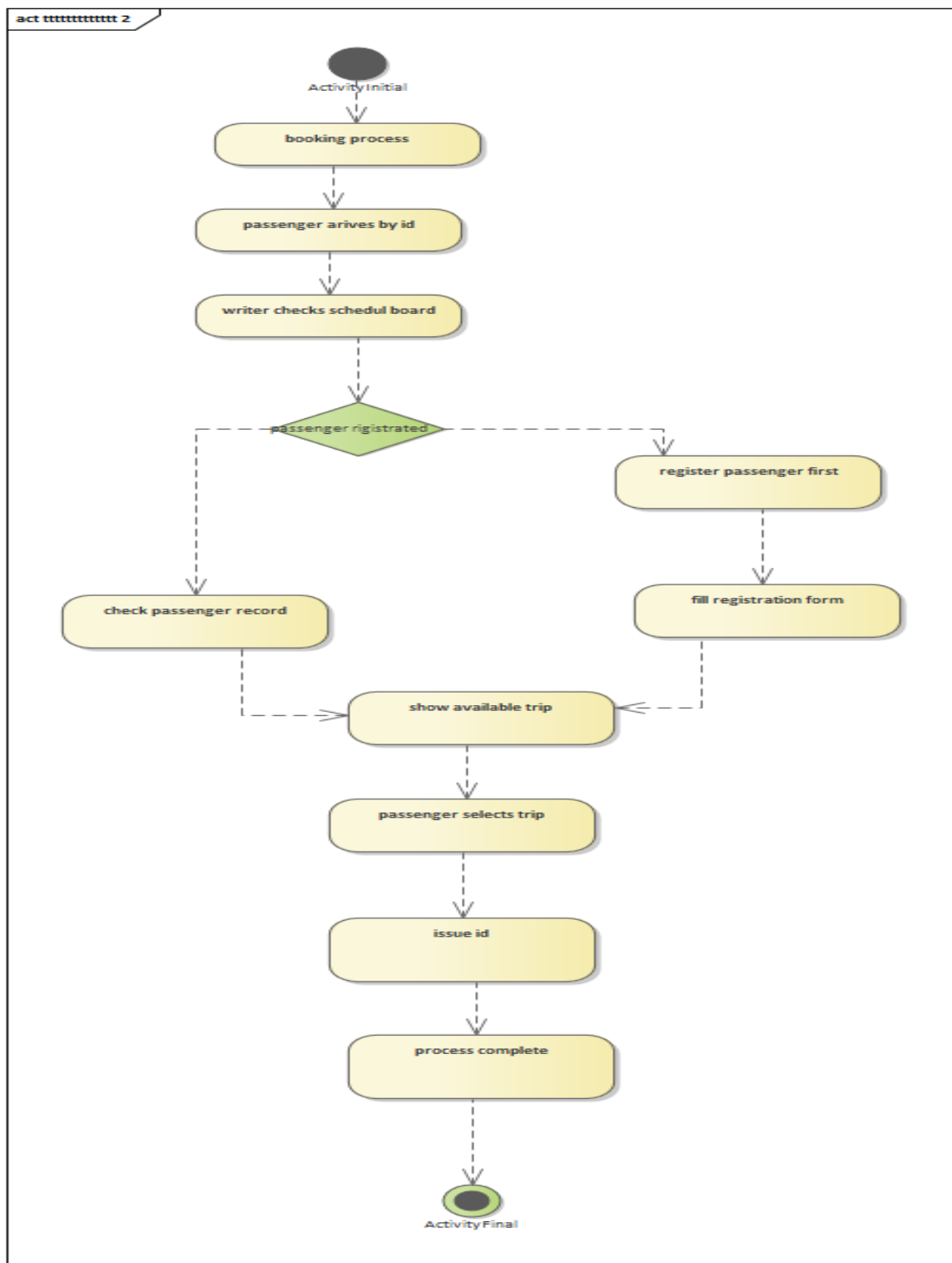


Figure 2.9 Activity Diagram (trip booking flow)

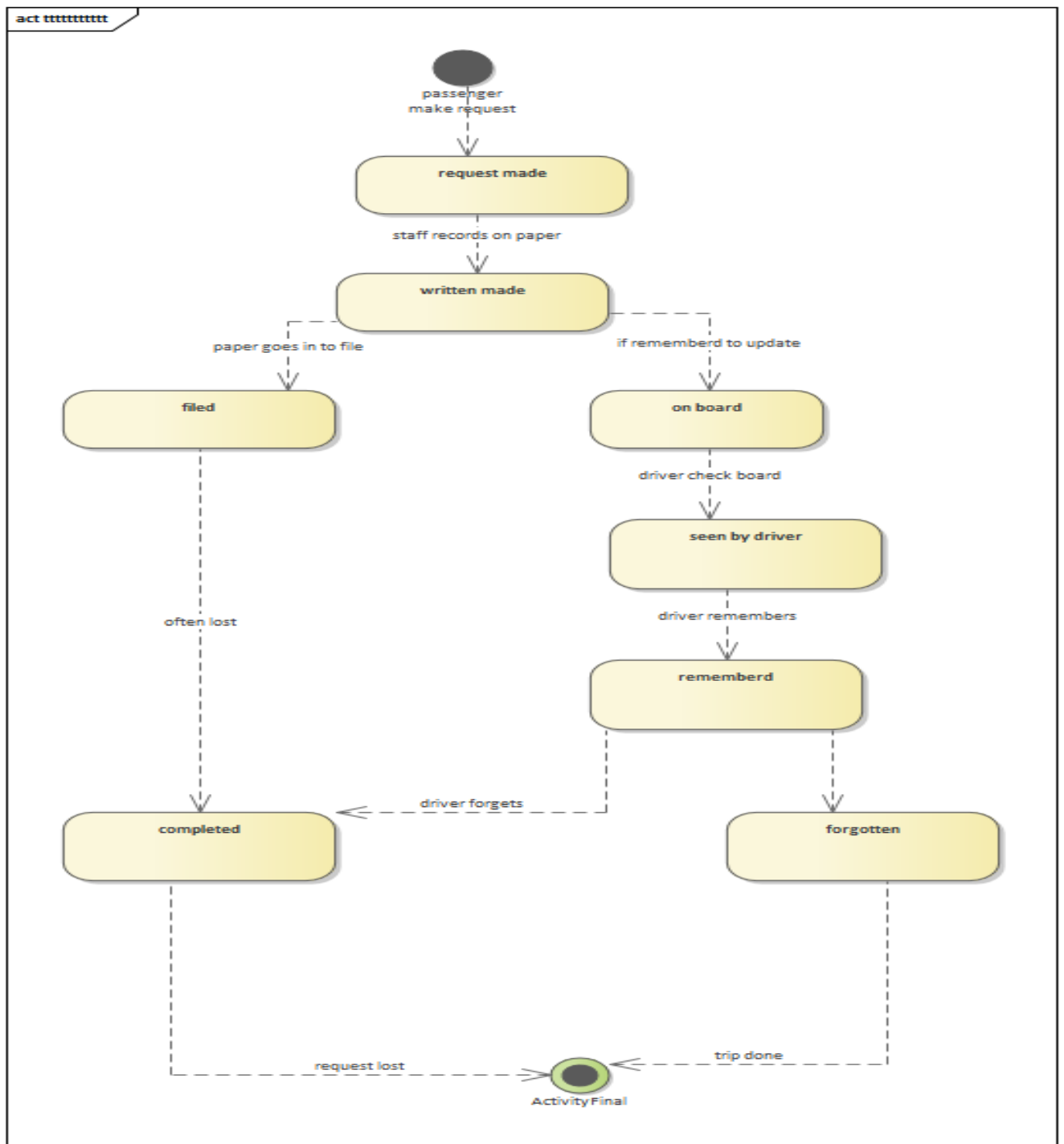


Figure 2.10 Activity Diagram(assign driver to vehicle flow)

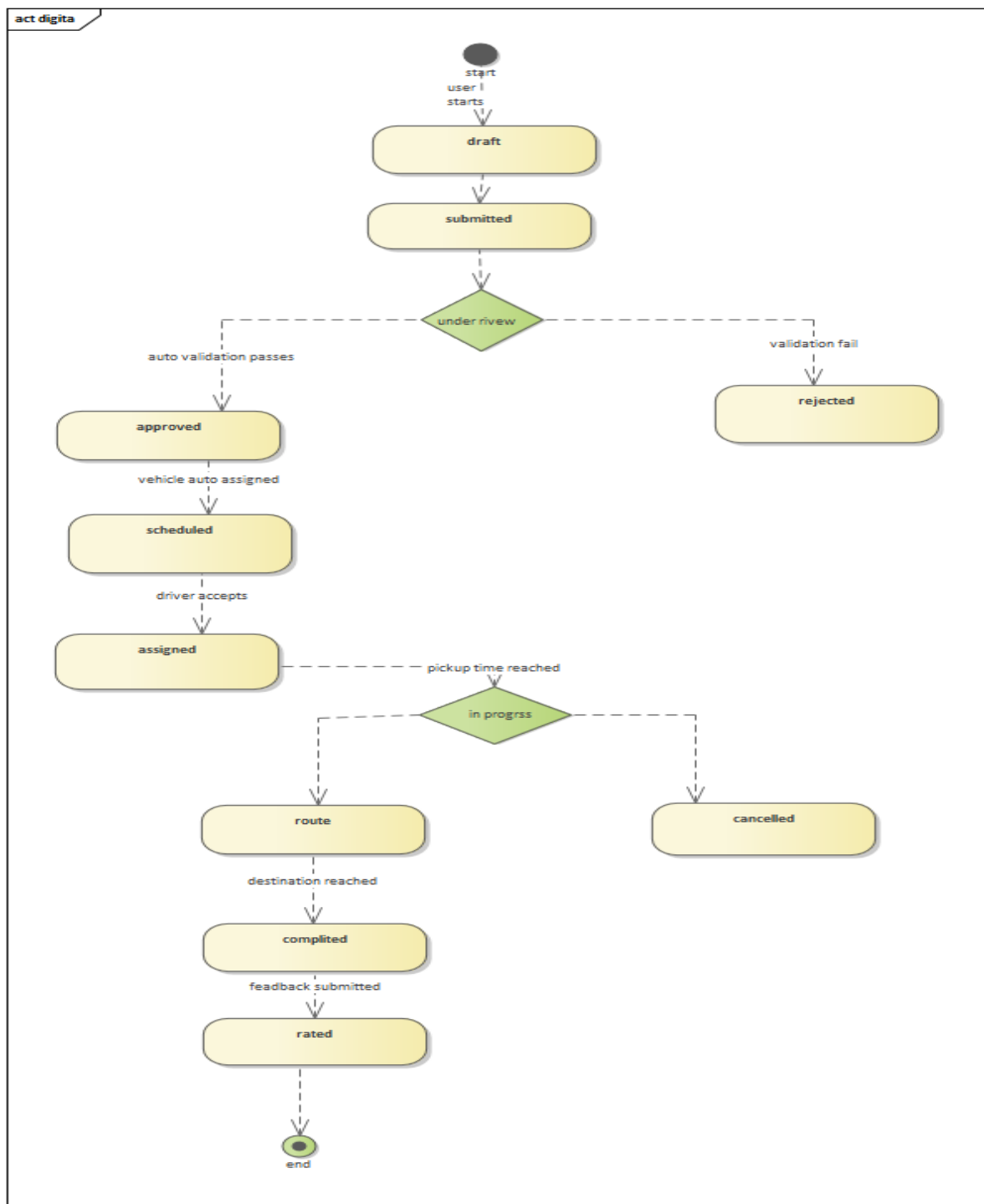


Figure 2.11 Activity Diagram(performance monitoring flow)

2.6.6.4 State Chart Diagram

2.6.6.5 Class Diagram

Shows classes like User, Vehicle, Driver, Request, etc., with their attributes.

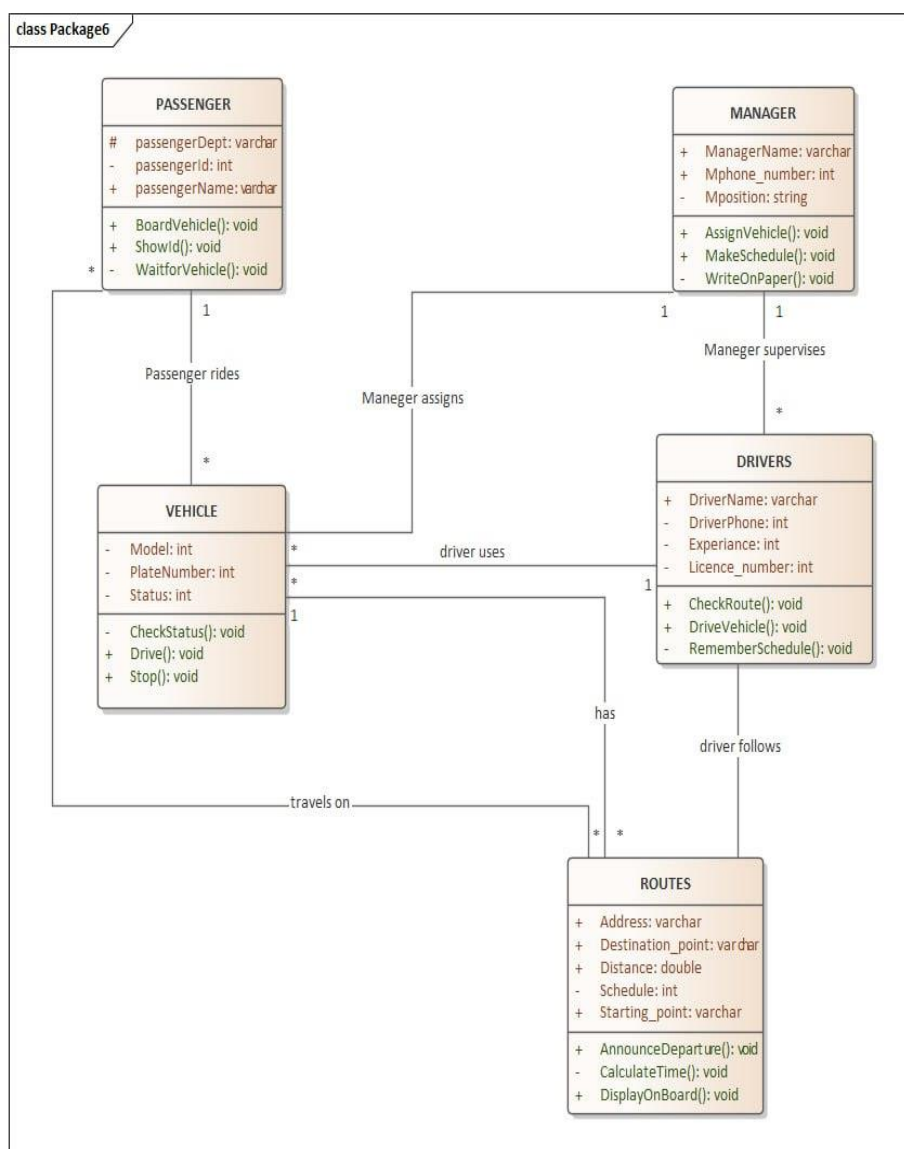


Figure 3.2 Class Diagram

Chapter Three: Database Design

3.1. Conceptual Database Design of the New System

The conceptual design for the Transportation System identifies the major data entities, their attributes, and the relationships among them. This design ensures that all transportation-related operations—such as scheduling buses, assigning drivers, booking trips, and processing systems—are logically represented.

3.1.1. Entities identification and Description

Below are the major entities for a Transportation System:

Entity	Description
# Passenger	Individuals who use the transportation service.
# Driver	Staff responsible for operating buses.
# Bus	Vehicles assigned for transportation.
# Route	Predefined paths the buses travel from origin to destination.
# Admin	System administrators who manage the entire transportation system.

3.1.2

Attribute Identification and Description

1. Passenger

PassengerID (PK) – unique identifier

FullName

PhoneNumber

Department

StudentID/StaffID (optional)

2. Driver

DriverID (PK)

FullName

LicenseNumber

PhoneNumber

ExperienceYears

3. Vehicle

PlateNumber(PK)

Model

Status (Active, Under Maintenance)

4. Route

RouteID (PK)

StartLocation

Destination

Distance

Address

-

5. Admin

Phone number

Fullname

Position

3.1.3 Relationship Identification and Description

Relationship	Type	Description
--------------	------	-------------

Passenger – Routine	M: M	A passenger travel on routine.
---------------------	------	--------------------------------

Routine _ Vehicle	1 : M	A Routine has vehicle.
-------------------	-------	------------------------

Manager_Vehicle	1 : M	one Manager supervisor many Vehicle.
-----------------	-------	--------------------------------------

Driver – Vehicle	1 : M	One driver can be assigned to many vehicles.
------------------	-------	--

Manager – Driver	1:M	one manager oversee many drivers.
------------------	-----	-----------------------------------

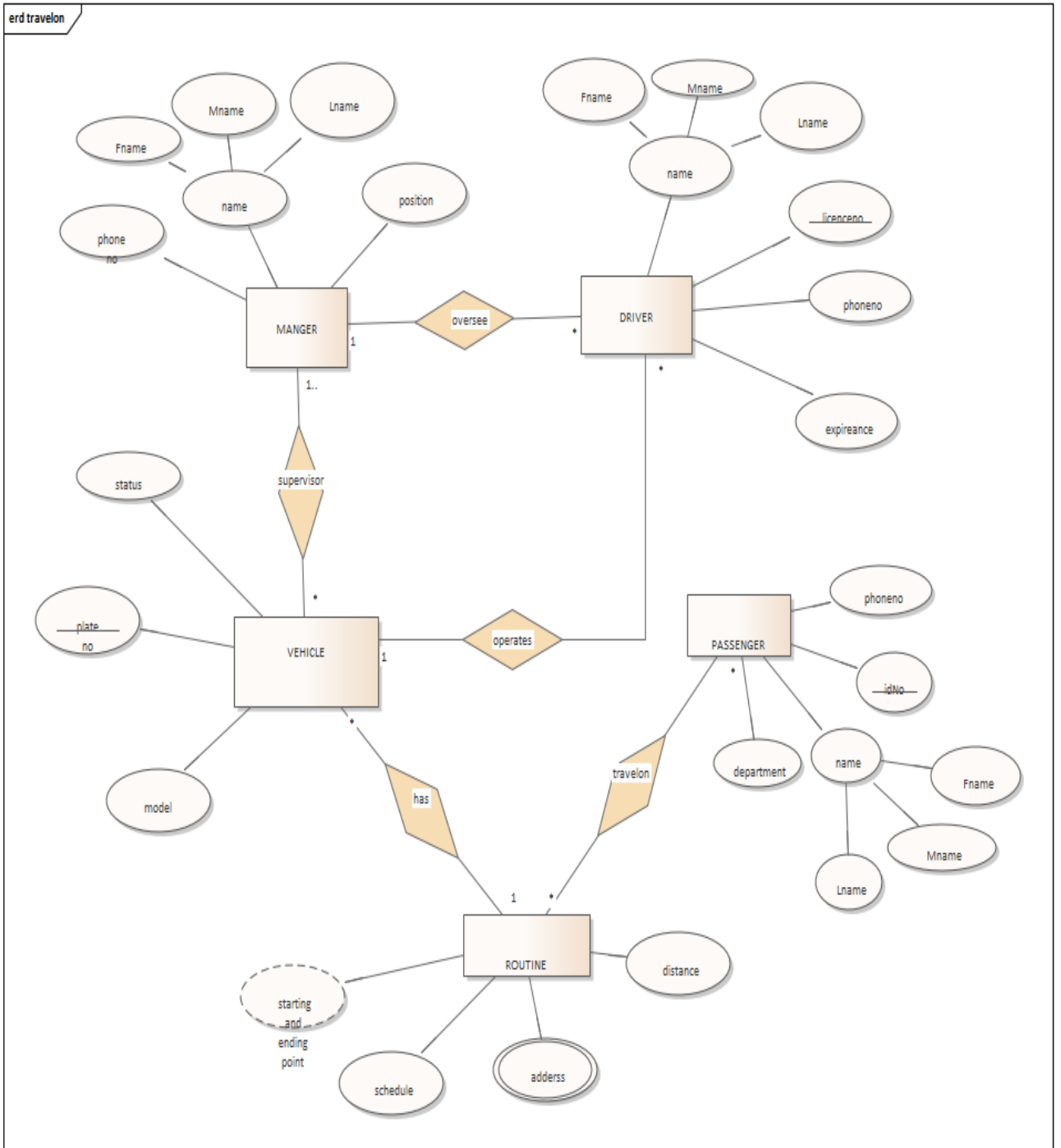


Figure 3.1 Entity Relationship Diagram